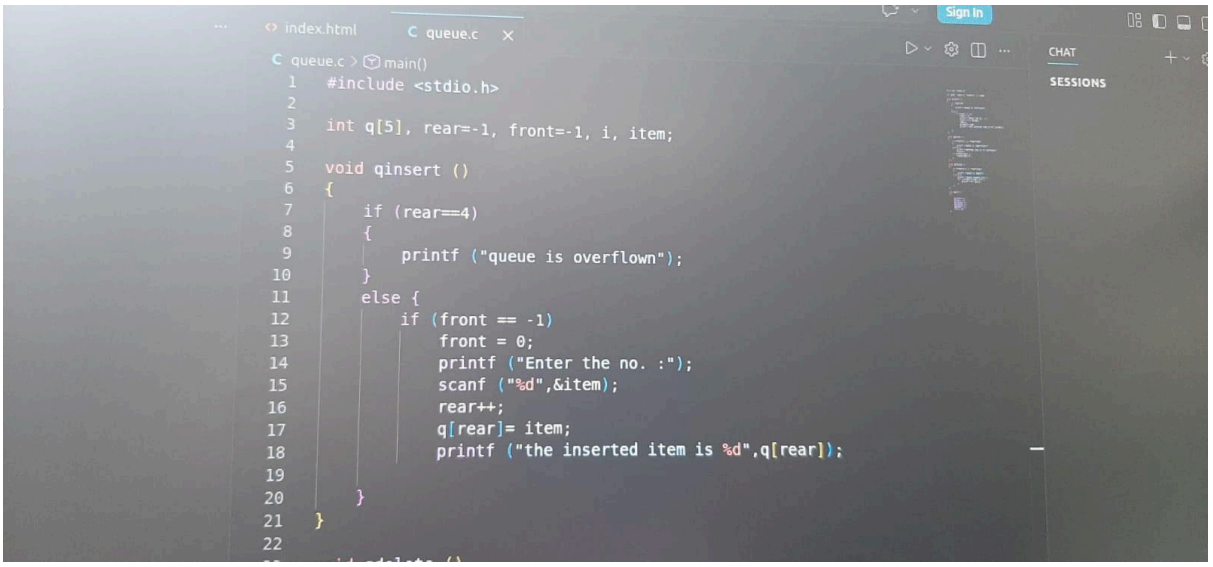
Aim : Implementation of queue using array

Program :

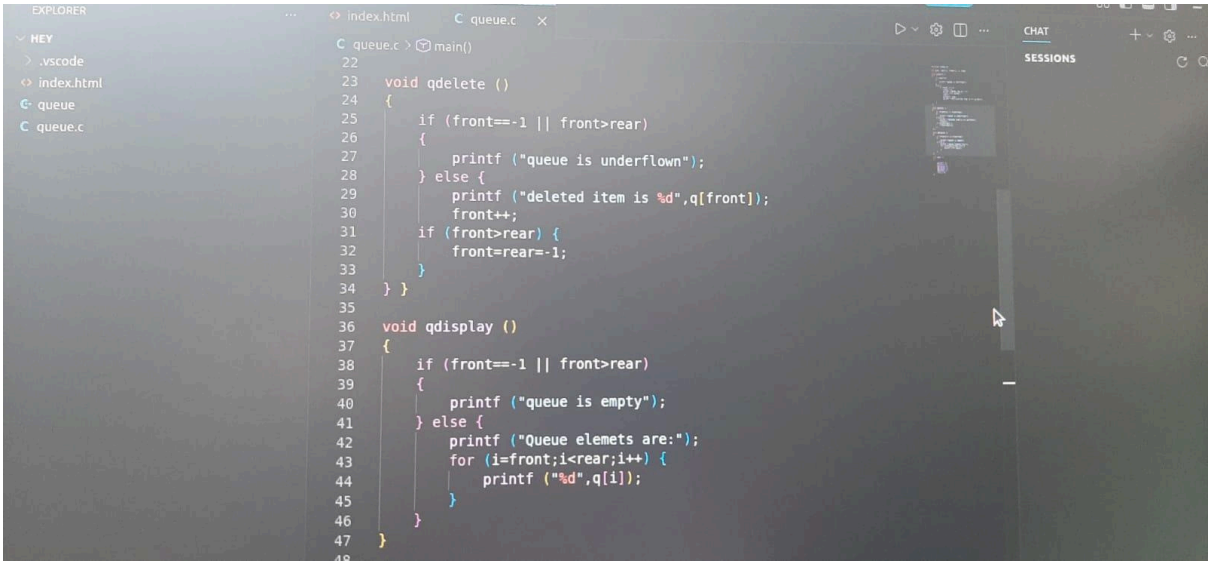


```c
#include <stdio.h>

int q[5], rear=-1, front=-1, i, item;

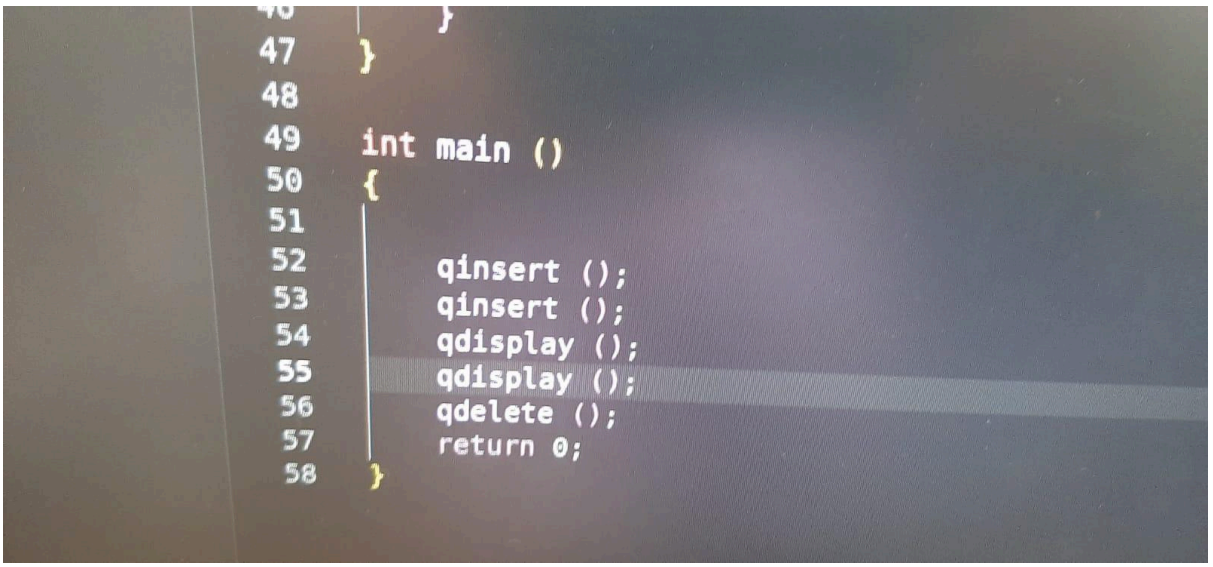
void qinsert ()
{
    if (rear==4)
    {
        printf ("queue is overflown");
    }
    else {
        if (front == -1)
            front = 0;
            printf ("Enter the no. :");
            scanf ("%d",&item);
            rear++;
            q[rear]= item;
            printf ("the inserted item is %d",q[rear]);

    }
}
```



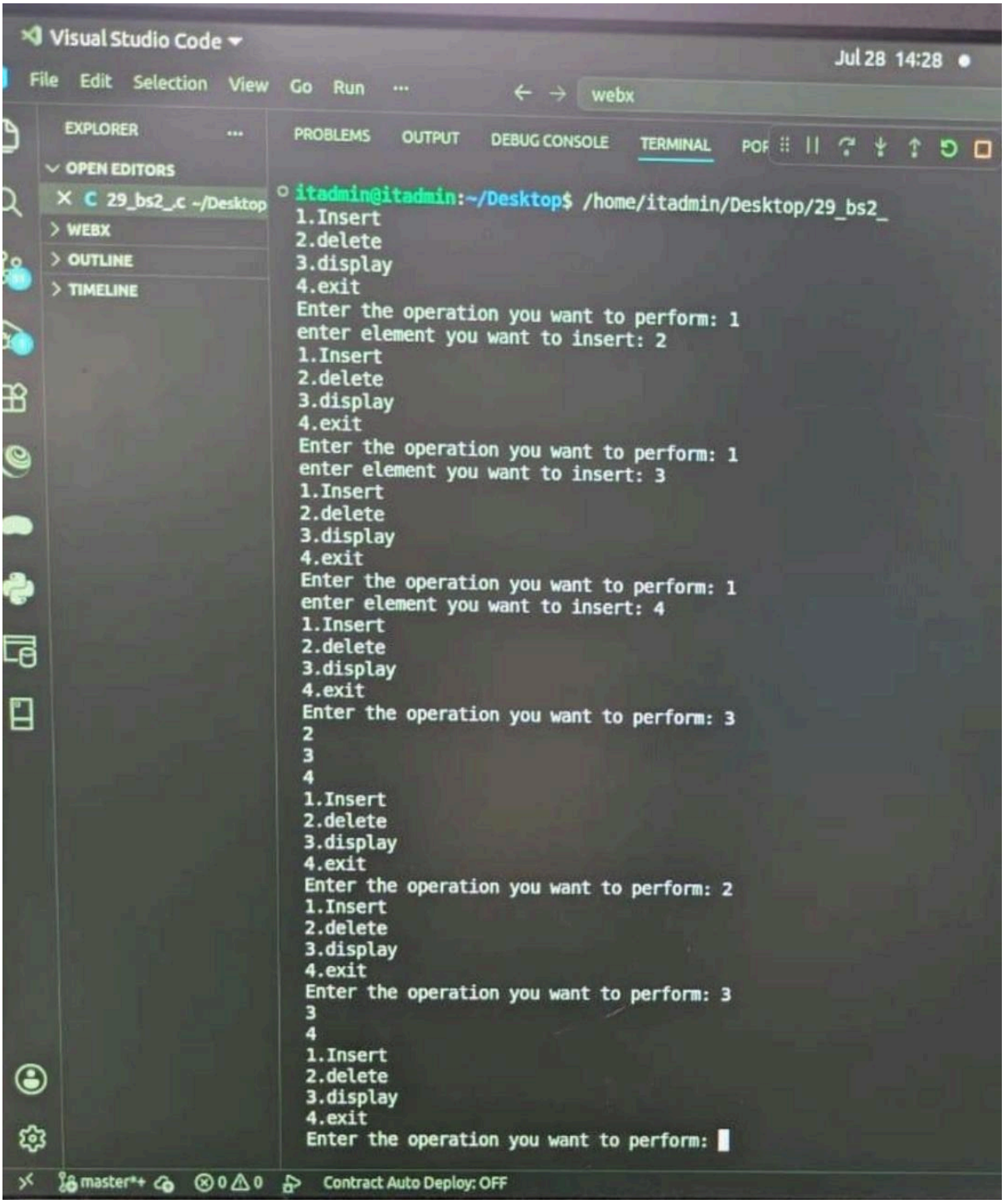
```c
void qdelete ()
{
    if (front==-1 || front>rear)
    {
        printf ("queue is underflown");
    } else {
        printf ("deleted item is %d",q[front]);
        front++;
    if (front>rear) {
        front=rear=-1;
    }
} }

void qdisplay ()
{
    if (front==-1 || front>rear)
    {
        printf ("queue is empty");
    } else {
        printf ("Queue elemets are:");
        for (i=front;i<rear;i++) {
            printf ("%d",q[i]);
        }
    }
}
```



```
47  }
48
49  int main ()
50  {
51
52      qinsert ();
53      qinsert ();
54      qdisplay ();
55      qdisplay ();
56      qdelete ();
57      return 0;
58  }
```

Output :


Visual Studio Code
Jul 28 14:28
File Edit Selection View Go Run ...
webx
EXPLORER
OPEN EDITORS
29_bs2_.c ~/Desktop
WEBX
OUTLINE
TIMELINE
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL
itadmin@itadmin:~/Desktop$ /home/itadmin/Desktop/29_bs2_
1.Insert
2.delete
3.display
4.exit
Enter the operation you want to perform: 1
enter element you want to insert: 2
1.Insert
2.delete
3.display
4.exit
Enter the operation you want to perform: 1
enter element you want to insert: 3
1.Insert
2.delete
3.display
4.exit
Enter the operation you want to perform: 1
enter element you want to insert: 4
1.Insert
2.delete
3.display
4.exit
Enter the operation you want to perform: 3
2
3
4
1.Insert
2.delete
3.display
4.exit
Enter the operation you want to perform: 2
1.Insert
2.delete
3.display
4.exit
Enter the operation you want to perform: 3
3
4
1.Insert
2.delete
3.display
4.exit
Enter the operation you want to perform:
master*+
0 0
Contract Auto Deploy: OFF